Problem Set 4

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1 Question 1

a The first regression implies that age is a determinant in explaining wage, while the second one would tell us the opposite. The relationship between how wage could determine age. I would say the first is more consistent with economic theory since we keep in mind that the relationship between wage and age is not that a higher wage implies that we magically become older but instead that as we get older our wage increases due to experience in the labor market.

b i The first one is true since as we can see $\frac{1}{.1172468} \neq .0110603$. This is mathematically done by,

$$\frac{wage_i - \alpha_i - \epsilon_i}{\beta_1} = age_i$$

However, if we look at what age is equal to

$$age_i = \alpha_2 + \beta_2 wage_i + \epsilon_i$$

Thus, for this to hold we would need the intercepts equal to zero as well as the error term. Which is an unreasonable assumption.

ii Since RSS is dependent on variance of its predictor then we can notice the following two fomrulas.

$$RSS_{wage} = (1 - r^2)var(wage)$$

$$RSS_{age} = (1 - r^2)var(age)$$

Thus since both of these depend on different varainces then they are clearly not going to be equal unless, the variances are equal. The rest, follow this idea for TSS and ESS.
For ESS

$$\hat{y}_{wage} \neq \hat{y}_{age}$$

thus when calculating it on the average input for it's own regression it will not equal each other. Especially since they are based on different errors thus they cant be equal.

iii Since R^2 is calculated as the proportion of variance explained, and the numerator involves the square of the covariance $\sum_i(x_i - \bar{x})(y_i - \bar{y})$, swapping x and y does not change this value. Mathematically, we have

$$R^2_{wage} = \frac{\hat{\beta}_1^2 \sum_i(age_i - a\bar{g}e)^2}{\sum_i(wage_i - w\bar{a}ge)^2} = \frac{\left[\sum_i(age_i - a\bar{g}e)(wage_i - w\bar{a}ge)\right]^2}{\sum_i(age_i - a\bar{g}e)^2 \sum_i(wage_i - w\bar{a}ge)^2} = R^2_{age}$$

Thus, the R^2 values are equal for both regressions.

iv The t-statistics for the slope coefficients are calculated by

$$t_{\hat{\beta}_1} = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}, \quad t_{\hat{\beta}_2} = \frac{\hat{\beta}_2}{SE(\hat{\beta}_2)}$$

Since in simple regression the t-statistic is related to R^2 by

$$t^2 = F = \frac{R^2}{1 - R^2} \cdot (n - 2),$$

and we already showed that $R^2_{wage} = R^2_{age}$, it follows that

$$t_{\hat{\beta}_1} = t_{\hat{\beta}_2}.$$

Thus, even though the slopes are different, the t-statistics are equal.

c That a big part of finding causal relationship is really understanding that x is the one that affects y and not the other way around. That the issue might be of reverse causality because we wouldn't say that your wage affects your age but instead that age is the true determinant of wage.